

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2023
CE 3151
(Fundamentals of Transportation Engineering)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume suitable data if necessary.

Section – A

1. (a) Define traffic zone. Enlist transport systems elements and types of transport planning. (10)
(b) How can a planner execute transport planning process? (06)
(c) Write down the methods of traffic volume study. How can a planner analyze transport demand? Explain in brief. (19)
2. (a) Define the following terms: (10)
i. ADT; ii. AWT; iii. AADT; iv. Daily variation factor; v. Monthly variation factor
(b) For Monday of a week, the value of yearly average traffic volume of a road is 2200 veh/day. This value is reduced by 200 veh/day every day before next Monday of that week. Calculate the daily adjustment factors for each day of that week. (10)
(c) What is O-D survey? What are the objectives of O-D survey and common methods for collecting O-D data? (15)
3. (a) Write down the types of parking facilities. Differentiate these types of parking facilities. (10)
(b) Explain the process of parallel parking with neat sketch. (12)
(c) What are field-based surveys? Explain in detail. (13)
4. (a) What is freight traffic survey? How can we overcome problems in goods movement? (08)
(b) Write down the importance of physical infrastructure for economic growth in terms of transportation planning. (05)
(c) Enlist the mode of transport. Write down the advantages and disadvantages of water transport. (12)
(d) What is hierarchy of roads? Describe the different types of roadway capacity according to O.K. Normann. (10)

Section – B

5. (a) What is sight distance? What are the factors that obstruct the sight distance? (08)
- (b) Derive an expression for finding stopping sight distance of road. (12)
- (c) A motorist travelling at 55 mi/h down a grade of 5% on a highway observes a crash ahead of him, involving an overturned truck that is completely blocking the road. If the motorist was able to stop his vehicle 30 ft from the overturned truck, what was his distance from the truck when he first observed the crash? Assume reaction time 2.5 sec. (15)
6. (a) Determine a safe overtaking sight distance for a design speed of 80 kmph. Assume data if necessary. (12)
- (b) Briefly discuss the sight distance at intersection. (08)
- (c) What is super elevation? Derive an expression for super elevation at level road. (15)
7. (a) Draw neat sketch of a typical urban and rural section with its major elements. (08)
- (b) A sag vertical curve is to be designed to join a -5% grade to +2% grade. If the design speed is 45 mi/h, determine the minimum length of the curve that will satisfy all criteria. Assume, $a=11.2 \text{ ft/sec}^2$ and perception reaction time = 2.5 sec. (20)
- (c) Why extra widening is provide on the horizontal curve? (07)
8. (a) Define the following terms: i. Kerb; ii. Camber; iii. Right of way; iv. Carriage way; v. PIEV (15)
- (b) Find the values of ruling minimum and absolute minimum radius of horizontal curve of a N.H. in plain terrain. Assume ruling design speed and minimum design speed values as 100 and 80 kmph⁻¹ respectively. (10)
- (c) Classify kerbs based on functions and describe briefly. (10)

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Examination, 2023

URP 3161
(Housing and Real estate Development)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) How would you define 'Need' and 'Demand'? "Housing can be both demand and need" – explain with your own justification. (10)
(b) With reference to the ecological system theory, explain how housing need is determined among different levels of environmental systems. (17)
(c) Explain the importance of housing for urban and regional planners. (08)
2. (a) Define slum as per the official definition of Bangladesh. Why do you think a significant portion of the urban population globally continue to live in slum? Explain. (08)
(b) Suppose, a person in Greenland slum earns a family income of 14000 BDT. They have two 2 school going children. They have an outstanding loan of 120,000 BDT with a monthly installment of 1500 BDT. In reference to the current Bangladesh's National Housing Policy, 2016, explain how that person can acquire a formal housing. Highlight both the short and long term household strategy. (18)
(c) "The poorer a person is, the longer distance the person travels" – explain in reference to Dhaka city. (09)
3. (a) Write a short note on the priorities of the National Housing Policy, 2016. In reference the subsequent policy adopted by the Bangladesh government, explain why or why not this housing this policy will make a significant contribution in the housing supply in Bangladesh. (18)
(b) "Bangladesh has missed major opportunities of increasing housing supply in the past"- how? (08)
(c) Singapore is a success story for homeownership. Highlight the major policy interventions by the Singapore government for increasing housing supply. (09)
4. (a) Write short notes on the following (any five): (35)
 - i. Housing supply and Demand
 - ii. Hedonic pricing
 - iii. Land Banking
 - iv. Sites and Services
 - v. Real-estate Investment Trust
 - vi. Condominium and Apartment
 - vii. Housing syndicate

Section – B

5. (a) Is real estate a property? What are the three fundamental ways to express real estate? Explain with examples. (20)
- (b) Real estate market tends to be highly segmented and localized.-Why? Explain with the relevant example. (15)
6. (a) Why investment in renovation and urban development is highly risky? Explain in terms of cost and cash flow uncertainty. Also explain the mitigation strategies to overcome the uncertainty. (15)
- (b) When will the downturn actually occur in a real estate market? Explain with supply and demand curve. How a developer will survive during a market downturn? (20)
7. (a) What is real estate valuation? Summarize the primary principles of real estate valuation. (20)
- (b) What are the adjustments need to be considered for valuation of property using the traditional sales comparison approach? (15)
8. (a) Write short notes on any five of the following: (35)
- i. Nonmonetary effects of investment
 - ii. Property cycles
 - iii. Supply side of urban growth
 - iv. Economic Base Multiplier
 - v. Bid-Rent curves
 - vi. Reproduction and replacement cost
 - vii. Accrued depreciation

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 3rd Year 1st Term Examination, 2023

URP 3121

(Rural Development Planning-I)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is Human Development Index (HDI)? Calculate HDI for the following countries and explain the results. (15)

Indicators	Country P	Country Q
Life Expectancy at Birth (Years)	74.5	70.5
Expected Years of Schooling (Years)	11.5	14.5
Mean Years of Schooling (Years)	7.0	8.5
Gross National Income per capita (2011 PPP \$)	13450	15240

- (b) Narrate the drawbacks of relying solely on GDP to measure rural development. (10)
- (c) What is depreciation? Describe the causes of depreciation. (10)
2. (a) What is GDP? Explain the components that are excluded in GDP calculation. (08)
- (b) Determine (i) Headcount Index (ii) Poverty Gap (PG) Index (iii) Squared Poverty Gap (SPG) Index for the following countries and explain the results. (20)
- Assume poverty line is 12000.

Country	Expenditures for each individual in countries					
A	11500	9500	16800	13000	8000	17400
B	16200	14000	9200	7000	11000	19100
C	6200	8500	9200	17500	13000	15500

- (c) What are the policy implications of different poverty measures? (07)
3. (a) What is Physical Quality of Life Index (PQLI)? Calculate PQLI for the following countries and explain the results. (15)

Country	Life Expectancy at age 1 (Year)	Infant Mortality (per 1000 live birth)	Literacy (Percent)
X	75	35	80
Y	65	28	85
Z	60	90	50

- (b) Make a comparative analysis between Nominal GDP and Real GDP with necessary examples. (12)
- (c) Write the advantages and disadvantages of Squared Poverty Gap (SPG) Index. (08)
4. (a) Suppose a society consists of 5 families whose incomes are \$16400, \$8300, \$28900, \$11200, \$21200. Construct a Lorenz curve for this five-family economy with quintile calculation. (15)
- (b) Write short notes on the following: (20)
- Poverty line
 - World Happiness Report
 - Gini Coefficient
 - Leading and Lagging indicator.



Section – B

5. (a) What is rural area? (05)
(b) Differentiate rural and urban community with appropriate examples. (10)
(c) Write note on the followings: (20)
 - i) Factors influence rural settlement pattern
 - ii) Scattered and gathered rural settlements.
6. (a) Briefly discuss the aspects of local raw material and skill-based industries with appropriate examples. (15)
(b) As a Rural Planner how will you identify a suitable rural industry based on industrial development screening matrix. Explain with appropriate examples. (20)
7. a) Write note on the followings: (20)
 - i) Bamboo walled and mud walled houses
 - ii) Timber and brick-built houses, and Tin sheet houses.
b) Discuss the level, radius of influence, service area, and population of different hierarchical rural centers. (15)
8. a) Provide a brief overview of the major components of "Comilla Model" in rural development of Bangladesh. (20)
b) Discuss the names and functions of government organizations and NGOs for the rural development of Bangladesh. (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2023
HUM 3117
(Accounting and Auditing for Planners)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Define accounting. Describe the various concepts of accounting. (10)
- (b) What is owner's equity? Discuss the elements of owner's equity along with their recording process in accounting. (10)
- (c) Define transaction and discuss the nature and characteristics of transaction. (10)
- (d) How to determine "Debit" and "Credit" of account. (05)

- 2 Firoz Kazi is a lawyer practicing in Dhaka Judge Court. He starts a sole proprietorship law firm in the name of Kazi Legal Services. During the first month of operation April 2023, Kazi engages in the following transactions.

2023	Kazi invested TK 2,00,000 cash to begin the firm.
April 1	
April 2	Purchased office supplies on account at a cost of TK 2,500.
April 3	Furniture purchased for office use TK 60,000 in cash.
April 4	Deals a case for one client and received cash TK 9,000.
April 5	Cash paid to creditor transaction April-2.
April 6	Advertisement bill received but not paid TK 5,000.
April 7	Utility bill received and paid in cash TK 500.
April 9	Pay TK 1,000 cash from personal fund for a personal trip to Rajshahi.
April 10	Kazi withdraw cash of TK 2,500 for personal use from the firm.
April 12	Cash paid for advertisement bill transaction April-6
Required:	
(a) Prepare Journal	
(b) Prepare a tabular summary and show the effects on accounting equation.	
Kazi uses the chart of accounts: Cash + supplies + furniture = Accounts payable + Kazi Capital. Other revenue expense and income are: Advertisement expense, Utility expense, Fees earned account.	

(20)
(15)

- 3 (a) From the following transactions you are required to prepare necessary ledger accounts. (35)

2023	Mr. Dulal invested cash Tk 195,000 in the business.
January 1	
January 2	Office rent paid in cash Tk 15,000.
January 3	Opened a bank account with cash Tk 40,000.
January 4	Office supplies purchased in cash Tk 4000.
January 6	Advertisement bill paid by cheque Tk 5000.
January 7	Service provided and bill to client TK 30,000.
January 8	Cash received from client transaction January 7.
January 10	Bank Loan taken in cash Tk 150,000 and signed a note.
Mr Dulal uses the following chart of accounts: Cash-101, Capital-301, Rent expense-201, Bank-110, Supplies-105, Advertisement expense-205, Service revenue-310, Accounts receivable-120, Notes payable-320.	



- 4 (a) Define Trial Balance. Describe the objectives of Trail Balance. (10)
- (b) Write down the limitations of Trial Balance. (05)
- (b) Mr. Azad has a trading concern owned only by him. The ledger account balances at the end of his accounting period ending June 30, 2023 are as follows: (20)

Particulars	Taka	Particulars	Taka
Azad Capital	5,81,700	Cash	52,250
Azad Drawing	17,560	Accounts receivable	72,000
Sales	290,060	Notes receivable	52,000
Sales return	4,450	Allowance for doubtful accounts	1,186
Purchases	176,950	Opening inventory	25,000
Purchase return	21,154	Office supplies	640

Repairs and Maintenances	11,680	Land	1,59,560
Office Salaries	41,060	Building	1,97,120
Advertisement expense	11,500	Accumulated depreciation building	39,424
Travel expenses	900	Office furniture	91,080
Prepaid insurance	1,700	Accumulated depreciation office furniture	22,770
Internet revenue	1,900	Accounts payable	71,168
Miscellaneous expense	650	Notes payable	31,778
Required: Prepare a Trial Balance as on 30 June 2023.			

Section – B

- 5 (a) Define audit. What are the elements of formal auditing? (10)
- (b) Briefly describe the objectives of auditing. (15)
- (c) Classify audit in terms of legal requirements. (10)
- 6 (a) What is engagement letter? Suppose you are an external independent auditor. Write a sample engagement letter for your new client, ABC Incorporation, a manufacturing company. (15)
- (b) Define internal control. How internal control help assessing different types of audit risks? Explain (10)
- (c) Define Audit Report. Discuss different types of audit report with example. (10)
- 7 (a) Explain the contents of company audit report, and reporting standards as prescribed by Institute of Chartered Accountants of Bangladesh. (15)
- (b) Briefly discuss any five standards on auditing as prescribed by International Standards on Auditing (ISA). (10)
- (c) State the provisions of reduced tax rates for income from export. What are the tax brackets for income from poultry firm, hatchery and fisheries? (10)
- 8 (a) Explain different types of non-NBR taxes with examples. (10)
- (b) Why taxation is necessary? (05)
- (c) Calculate income tax payable for an individual assessee as per latest income tax act and related rules: (20)

Particulars	Amount (Tk)
Basic Salary	70,000/month
Special Pay	30,000 (total)
Dearness allowances	@ 10% of basic
House rent	@ 60% of basic
Medical allowance	1500/month
Festival allowances	140,000 (total)
Bengali new year allowance	14,000 (total)
Conveyance allowance	50,000 (total)
Overtime allowance	50,000 (total)
Bonus	80,000 (total)
Interest Earned: Provident fund	20,000 (total)
Deductions and Investments:	
Contribution to provident fund	@ 10% of basic
Contribution to group insurance and benevolent fund	7,000
Life insurance premium	50,000
Deposit pension scheme	60,000
Contribution to Zakat fund	200,000
Investment in listed securities	500,000
Income tax deducted at source (Income Area: Dhaka City Corporation)	84,000
Required: (i) Calculate total tax liability. (ii) Prepare a schedule 24-A (Income from Salary). (iii) Show calculations of allowable investment.	

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
 BURP 3rd Year 1st Term Examination, 2023.
CE 3107
 (Elements of Solid Mechanics)

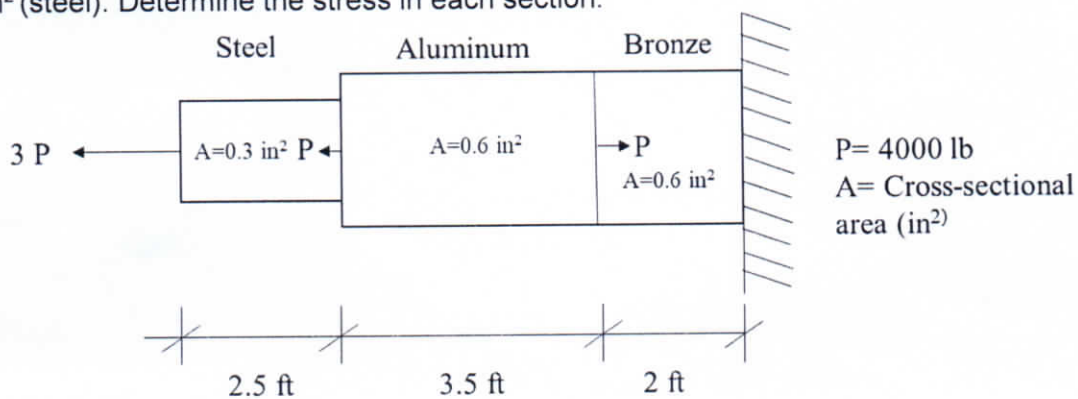
Full Marks: 210

Time: 3.0 hrs

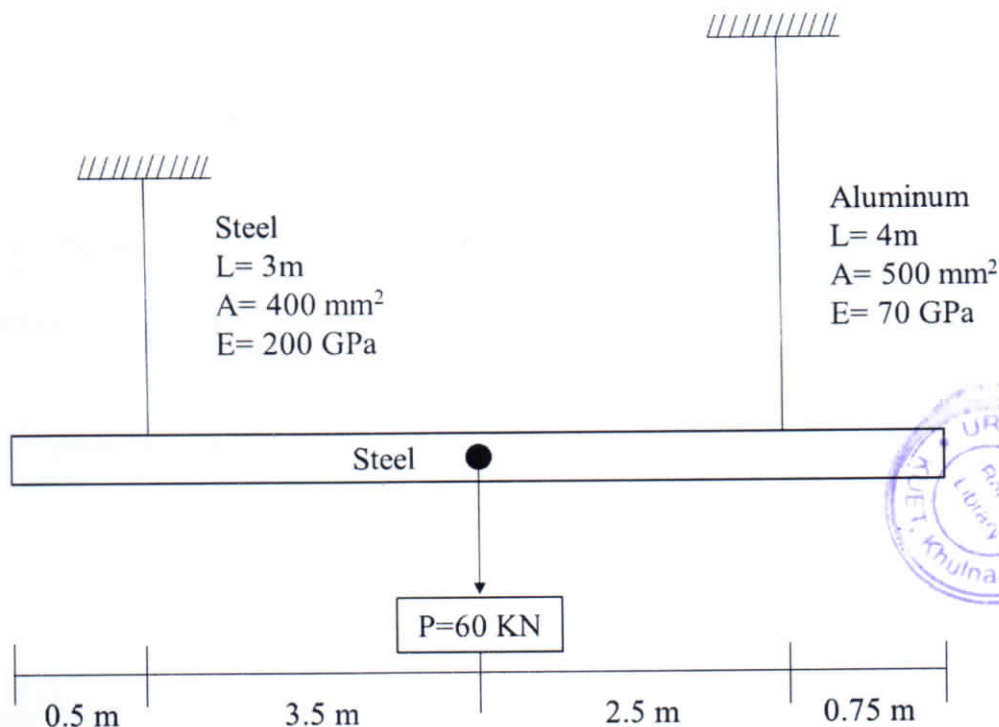
- N.B.: i) Answer any **three** questions from each section in separate script
 ii) Figures in the right margin indicate marks
 iii) Assume reasonable standard value for any missing data.

Section - A

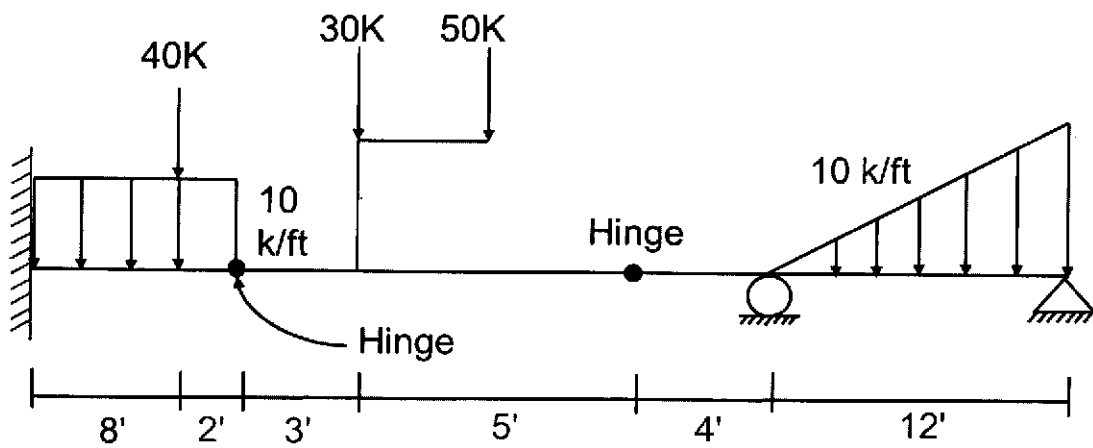
1. (a) Define (i) elastic limit (ii) rupture strength (iii) Necking. (06)
 (b) Derive the following formula, $\delta = PL/AE$ where symbol bears their usual meaning. (06)
 (c) A rod is composed of an aluminium section rigidly attached between steel and bronze sections, as shown in figure below. The cross sectional area of rod is 0.6 in^2 and 0.3 in^2 (steel). Determine the stress in each section. (11)



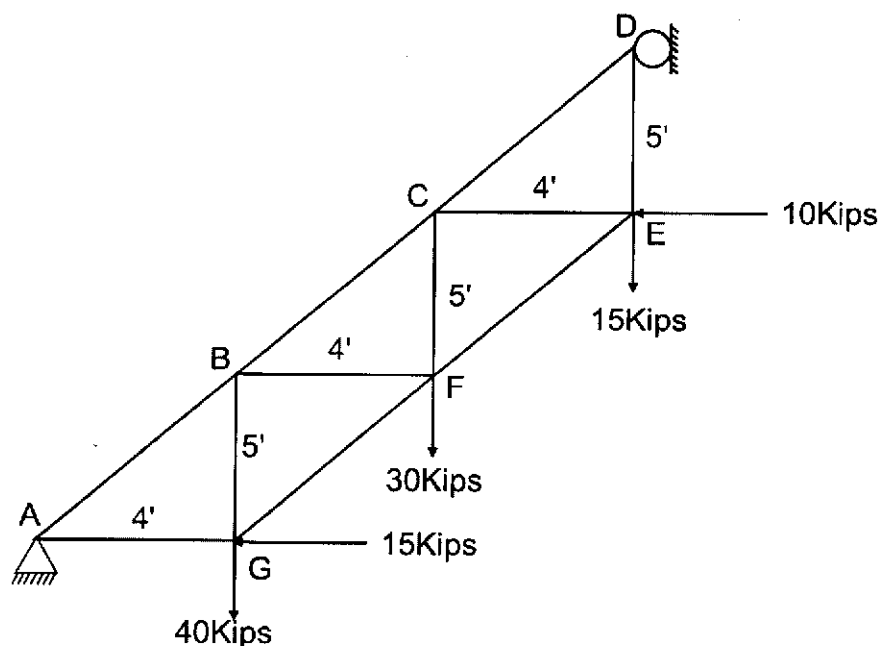
- (d) The rigid bar AB, attached to two vertical rod as shown in figure below, is horizontal before the load is applied. Determine the vertical movement of P if its magnitude is 60 KN. (12)



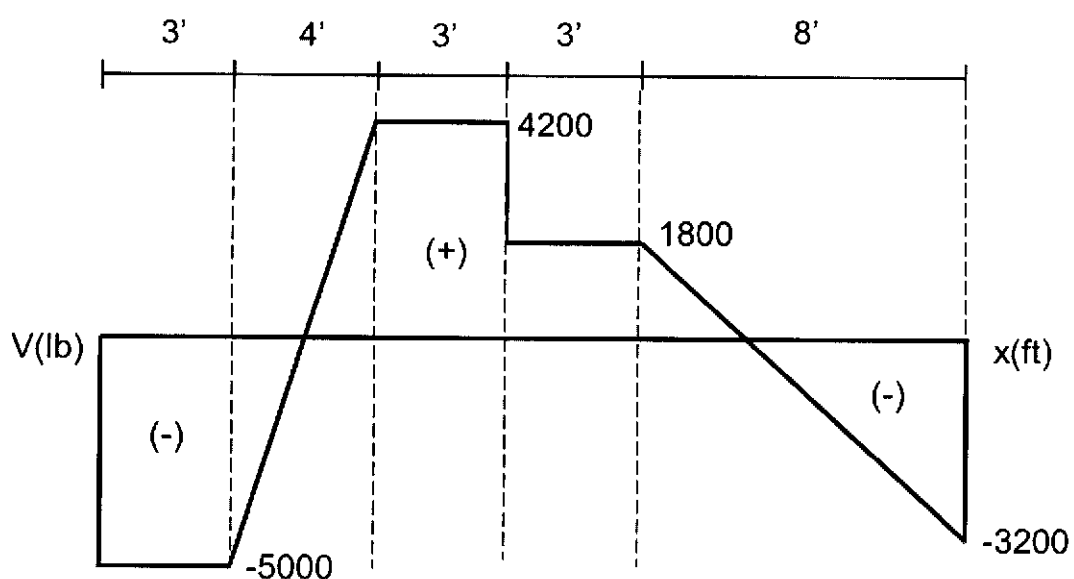
2. (a) Define Shear Force and Bending Moment. Establish the relationship between load, shear and moment. (10)
 (b) Draw the shear force diagram and bending moment diagrams for the beam shown in figure below. (25)



3. (a) Draw stress-strain diagram of mild steel and briefly discuss its application in Reinforced concrete structure. (10)
- (b) Determine the cross-sectional areas of selected member BC, CD, BG, FC, GF. For the truss shown below. The stresses are not exceed 25 ksi in tension or 15 ksi in compression. (25)



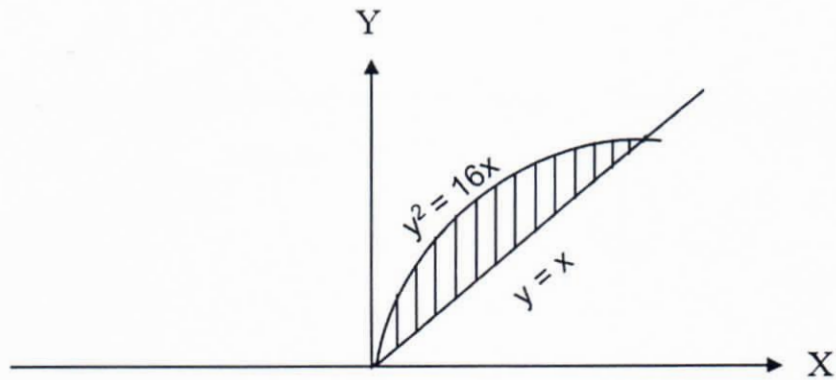
4. (a) What is meant by flexural formula? Prove that flexural stress in any section varies with the distance of the section from the neutral axis. (12)
- (b) In the following beam, draw the moment and load diagram to the given shear diagrams. Specify at all change of load and at all points of zero shear. (15)



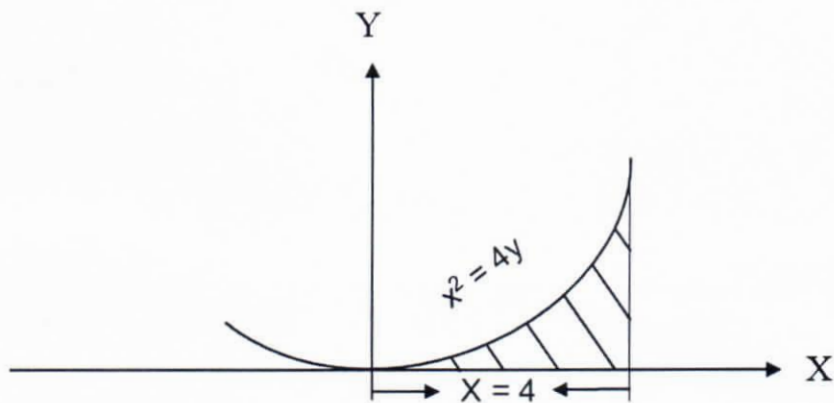
- (c) What are the different types of support and loading? Write down the sign convention of shear force and bending moment with neat sketches. (08)

Section – B

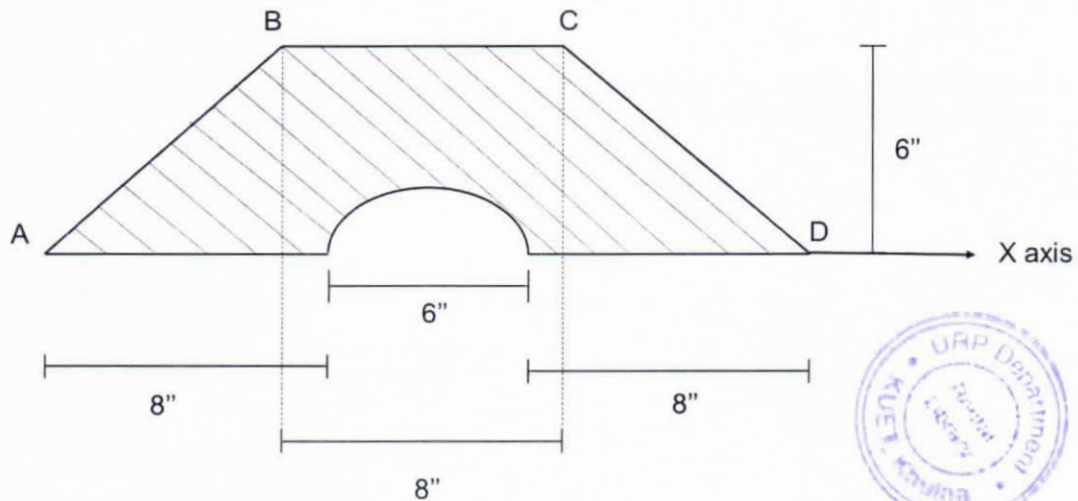
5. (a) What is meant by centroid and center of gravity? Determine the location of the centroid of a plane triangle. (12)
- (b) Find the coordinates of the centroid of the area bounded by the curves $y^2 = 16x$ and $y = x$ as shown in figure below. (23)



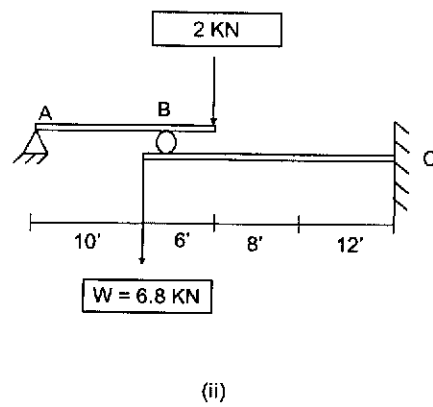
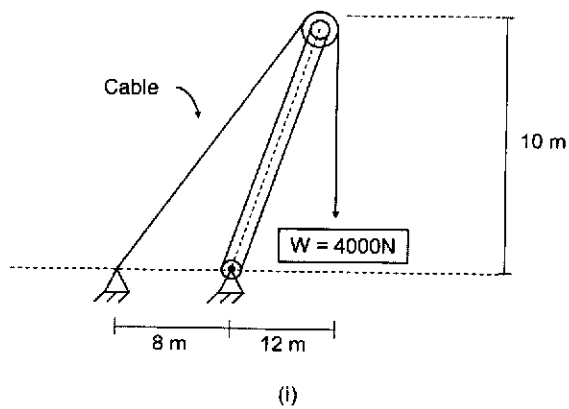
6. (a) For the shaded area in figure, by direct integration method find (i) I_x (ii) I_y (18)



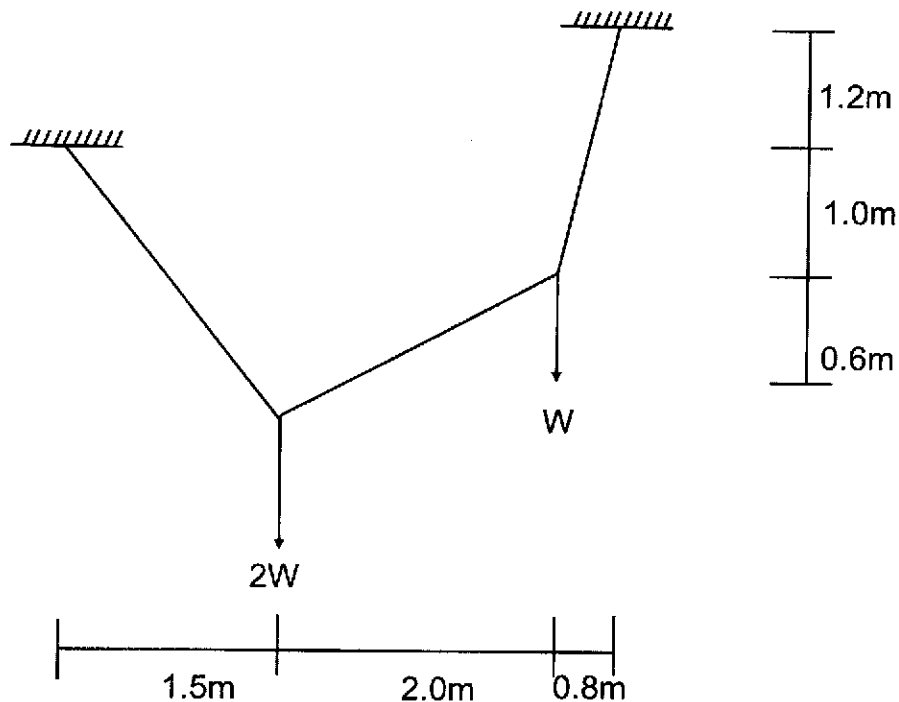
- (b) Determine the I_x for the shaded area shown in figure below (17)



7. (a) What is transmissibility of force? State the principle of transmissibility states with suitable example. (13)
- (b) Each of the concurrent force is in static equilibrium as shown: (22)
- (i) Construct the free body diagram of each problems.
- (ii) Solve for the unknown forces to maintain static equilibrium.



8. (a) For a given structure shown in figure, determine the safe load "W" that can be applied to the structure. (18)



Member	Diameter	Allowable Stress
AD	80 mm	120 MPa
AB	75 mm	80 MPa
BC	50 mm	60 MPa

- (b) Two spheres are at rest against smooth surfaces as shown below. Sphere A weighs 5000 lb and B weighs 1500 lb. Find the reaction at C, D and E. (17)

